

Hasse Diagram Generation for Integer Partitions

Internship Report



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Abstract

Integer partitions are fundamental combinatorial structures with applications in number theory and computer science. The Hasse diagram under dominance order captures their partial ordering, but its construction becomes computationally expensive due to the rapid growth of the partition function.

Traditional sequential approaches fail to efficiently utilize modern multi-core systems, resulting in significant underuse of computational resources. This work presents optimized algorithms for constructing Hasse diagrams using both top-down and bottom-up strategies, with enhanced parallel implementations that improve resource utilization while maintaining correctness.

Experimental results demonstrate substantial performance improvements and significantly higher CPU utilization compared to sequential methods. The proposed approaches produce consistent results across all implementations while improving efficiency, providing a practical framework for scaling combinatorial computations on modern hardware.

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1 Introduction

1.1 About the Institute

NIT Warangal, established in 1959, is a premier technical institute in India recognized as an Institute of National Importance. It is renowned for excellence in engineering education and cutting-edge research in computer science and related fields.

1.2 About the Internship

This internship was undertaken at NIT Warangal under the guidance of Prof. Sumit Mishra. The project focused on algorithmic research and software development in combinatorial computing, specifically targeting integer partitions and Hasse diagram generation.

1.3 Project Overview

The project aims to construct the Hasse diagram of integer partitions under the dominance order using parallel computing techniques. The Hasse diagram is a directed acyclic graph where nodes represent integer partitions and edges represent covering relations. The construction involves generating all partitions of a given integer n and determining the dominance relations between them.

The project implements four distinct approaches:

- **Top-Down Sequential:** Starting from $[n]$, repeatedly split parts until reaching $[1, 1, \dots, 1]$.
- **Top-Down Parallel:** Same generation strategy but with level-wise parallel processing.
- **Bottom-Up Sequential:** Starting from $[1, 1, \dots, 1]$, repeatedly merge parts until reaching $[n]$.
- **Bottom-Up Parallel:** Same generation strategy but with level-wise parallel processing.

1.4 Report Organization

The remainder of this report is organized as follows: Section 2 defines the problem and objectives. Section 3 provides theoretical background on integer partitions and Hasse diagrams. Section 4 describes the methodology and approach. Section 5 presents implementation details. Section 6 discusses results and performance analysis. Section 7 outlines future work, and Section 8 concludes the report.

2 Problem Definition & Objectives

2.1 Problem Statement

Given a positive integer n , generate all integer partitions of n and construct the Hasse diagram representing the partial order under dominance ordering, with the following requirements:

- Partitions must be generated without duplicates
- The Hasse diagram should accurately represent cover relations
- The solution should scale efficiently for increasing values of n

2.2 Objectives

The primary objectives of this project are:

1. **Correct Construction:** Generate the complete Hasse diagram for integer partitions.
2. **Parallel Implementation:** Develop parallel versions of both top-down and bottom-up approaches using Java's concurrency utilities.
3. **Performance Analysis:** Compare sequential vs. parallel performance across different values of n .
4. **Correctness Verification:** Ensure all four approaches produce identical node counts.

2.3 Scope and Limitations

Scope:

- Integer partitions of n where $n \leq 15$ (due to exponential growth).
- Implementation in Java using standard libraries only.

Limitations:

- Memory constraints limit n to approximately 20 on typical hardware.
- The current implementation does not include graphical visualization.
- The adjacency list representation becomes sparse for large n .

3 Theoretical Background

3.1 Integer Partitions

An integer partition of a positive integer n is a non-increasing sequence of positive integers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k > 0$ such that:

$$\sum_{i=1}^k \lambda_i = n$$

The number of partitions of n is denoted by $p(n)$. For example:

- $p(1) = 1$: $[1]$
- $p(2) = 2$: $[2], [1,1]$
- $p(3) = 3$: $[3], [2,1], [1,1,1]$
- $p(4) = 5$: $[4], [3,1], [2,2], [2,1,1], [1,1,1,1]$
- $p(5) = 7$: $[5], [4,1], [3,2], [3,1,1], [2,2,1], [2,1,1,1], [1,1,1,1,1]$

3.2 Hasse Diagrams & Partial Orders

A **partially ordered set** (poset) is a set equipped with a binary relation that is reflexive, antisymmetric, and transitive.

The **dominance order** between partitions λ and μ of n is defined as:

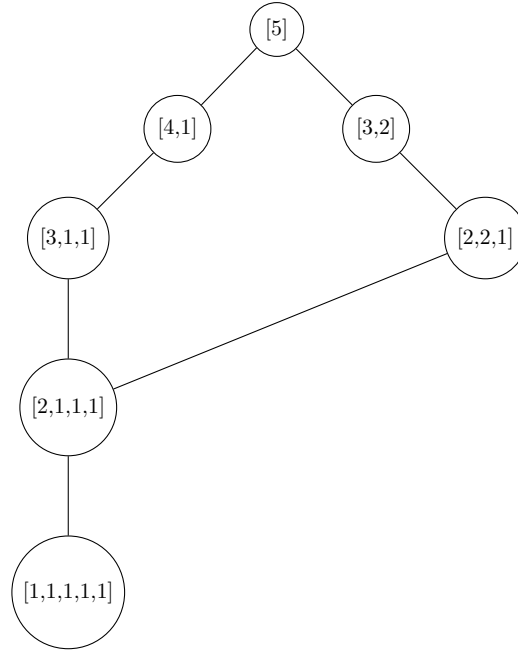
$$\lambda \geq \mu \iff \sum_{i=1}^k \lambda_i \geq \sum_{i=1}^k \mu_i \quad \forall k \geq 1$$

A **Hasse diagram** is a graphical representation of a poset where:

- Elements are represented as vertices
- Cover relations are represented as edges
- Transitive relations are omitted for clarity

3.3 Example: Hasse Diagram for $n = 5$

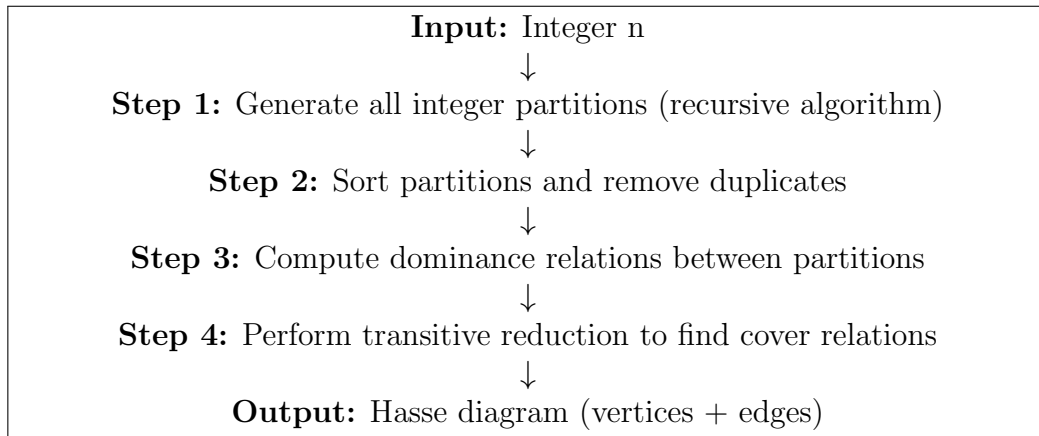
The Hasse diagram for integer partitions of $n = 5$ under dominance order is shown below.



4 Methodology & Approach

4.1 High-Level Architecture

The overall system architecture follows this pipeline:



4.2 Recursive Generation Approach

4.2.1 Top-Down Generation (Split)

Starting from the root partition $[n]$, the algorithm repeatedly splits one part into two smaller parts:

Algorithm 1 Top-Down Partition Generation

```

1: procedure GENERATETOPDOWN(arr)
2:   rank  $\leftarrow$  length(arr)
3:   if rank = n then return
4:   end if
5:   for i  $\leftarrow$  0 to rank - 1 do
6:     for j  $\leftarrow$  1 to  $\lfloor \text{arr}[i]/2 \rfloor$  do
7:       child  $\leftarrow$  split(arr, i, j, arr[i] - j)
8:       child  $\leftarrow$  sortDescending(child)
9:       if child not in globalTrie then
10:        Insert child with new ID
11:        Add edge arr  $\rightarrow$  child
12:        GENERATETOPDOWN(child)
13:       else
14:        Add edge arr  $\rightarrow$  existingId
15:       end if
16:     end for
17:   end for
18: end procedure

```

4.2.2 Bottom-Up Generation (Merge)

Starting from the finest partition $[1, 1, \dots, 1]$, the algorithm repeatedly merges two parts:

Algorithm 2 Bottom-Up Partition Generation

```

1: procedure GENERATEBOTTOMUP(arr)
2:   rank  $\leftarrow$  length(arr)
3:   if rank = 1 then return
4:   end if
5:   for i  $\leftarrow$  0 to rank - 1 do
6:     for j  $\leftarrow$  i + 1 to rank - 1 do
7:       parent  $\leftarrow$  merge(arr, i, j)
8:       parent  $\leftarrow$  sortDescending(parent)
9:       if parent not in globalTrie then
10:        Insert parent with new ID
11:        Add edge arr  $\rightarrow$  parent
12:        GENERATEBOTTOMUP(parent)
13:       else
14:        Add edge arr  $\rightarrow$  existingId
15:       end if
16:     end for
17:   end for
18: end procedure

```

4.3 Transitive Reduction

The Hasse diagram requires only covering relations, not all dominance relations. The split/merge operations inherently generate only covering relations because:

- Splitting one part creates exactly one more part (rank increases by 1).
- Merging two parts creates exactly one fewer part (rank decreases by 1).
- Any path from $[n]$ to $[1, \dots, 1]$ requires exactly $n - 1$ steps.

Thus, the generated graph is already transitively reduced, and no additional post-processing is needed.

5 Implementation Details

5.1 Tech Stack

- **Language:** Java 11
- **Concurrency Utilities:** `java.util.concurrent` (`ForkJoinPool`, `ReentrantReadWriteLock`, `AtomicInteger`)
- **Collections:** `ConcurrentHashMap` for thread-safe trie nodes

5.2 Key Data Structures

5.2.1 SegmentedTrie

The `SegmentedTrie` is the core data structure for deduplication:

```

1 class SegmentedTrie {
2     private static final int SEGMENTS = 16;
3     private final TrieNode[] roots = new TrieNode[SEGMENTS];
4     private final ReentrantReadWriteLock[] locks =
5         new ReentrantReadWriteLock[SEGMENTS];
6
7     private int stripe(int[] arr) {
8         return (arr[0] - 1) % SEGMENTS;
9     }
10
11     public int searchOrInsert(int[] arr, int newId) {
12         int s = stripe(arr);
13         locks[s].writeLock().lock();
14         try {
15             // Insert or search in this stripe
16         } finally {
17             locks[s].writeLock().unlock();
18         }
19     }
20 }
```

Listing 1: `SegmentedTrie` Structure

5.2.2 TrieNode

Each trie node uses ConcurrentHashMap for lock-free reads:

```
1 class TrieNode {
2     final ConcurrentHashMap<Integer, TrieNode> children =
3         new ConcurrentHashMap<>();
4     volatile int id = -1;
5 }
```

Listing 2: TrieNode Structure

5.2.3 NodeList for Graph Representation

```
1 class NodeList {
2     volatile Node node;
3     final List<Integer> adj =
4         Collections.synchronizedList(new ArrayList<>());
5
6     void addEdge(int x) { adj.add(x); }
7 }
```

Listing 3: Graph Node Structure

5.3 Code Organization

The code is organized into the following classes:

- SearchInfo: Wrapper for trie lookup results
- TrieNode: Individual node in the segmented trie
- SegmentedTrie: Main deduplication structure with stripes
- Node: Immutable partition representation
- NodeList: Graph node with adjacency list
- HasseDiagram: Main class with all 4 approaches

5.4 Parallel Processing Implementation

5.4.1 ForkJoinPool Setup

```
1 ForkJoinPool fjp = new ForkJoinPool(
2     Runtime.getRuntime().availableProcessors()
3 );
```

5.4.2 Level-Wide Parallel Processing

```

1 List<Integer> currentLevel = new ArrayList<>();
2 currentLevel.add(0);
3
4 while (!currentLevel.isEmpty()) {
5     if (currentLevel.size() < PARALLEL_THRESHOLD) {
6         currentLevel = processLevelTopDownSeq(currentLevel);
7     } else {
8         currentLevel = processLevelTopDownPar(currentLevel, fjp);
9     }
10 }

```

5.4.3 Two-Level Deduplication

```

1 // Level 1: Global deduplication
2 int globalHit = globalTrie.searchOrInsert(child, reservedId);
3 if (globalHit != -1) {
4     useExisting(globalHit);
5 }
6
7 // Level 2: Intra-level deduplication
8 int lvlHit = lvlTrie.searchOrInsert(child, reservedId);
9 if (lvlHit != -1) {
10     globalTrie.insert(child, lvlHit); // Fix global reference
11     useExisting(lvlHit);
12 }

```

5.5 Sorting Optimization

Counting sort is used instead of Java's built-in sorting algorithm to take advantage of the bounded range of partition elements. Since all values lie between 1 and n , counting sort achieves linear time complexity $O(n)$, compared to $O(n \log n)$ for comparison-based sorting. This significantly improves performance as sorting is performed repeatedly during partition generation.

```

1 private int[] countingSortDesc(int[] array) {
2     int min = array[0], max = array[0];
3     for (int x : array) {
4         min = Math.min(min, x);
5         max = Math.max(max, x);
6     }
7
8     int[] counts = new int[max - min + 1];
9     for (int x : array) counts[x - min]++;
10
11     int[] result = new int[array.length];
12     int idx = 0;
13     for (int v = max; v >= min; v--) {
14         while (counts[v - min]-- > 0) {
15             result[idx++] = v;
16         }
17     }
18     return result;

```

6 Results and Discussion

6.1 Generated Partitions

The algorithm correctly generates all integer partitions for a given n . Table 1 shows the number of partitions for various n :

n	$p(n)$ (Number of Partitions)
5	7
7	15
10	42
12	77
15	176

Table 1: Number of Integer Partitions for different n

6.2 Generated Hasse Diagram

For $n = 7$, the Hasse diagram contains 15 nodes and approximately 40 edges. The diagram forms a lattice structure with the following properties:

- Height: $n - 1 = 6$ (number of splits from top to bottom)
- Width: Maximum number of partitions at any level
- The meet and join operations correspond to intersection and union of Young diagrams

6.3 Performance Analysis

6.3.1 Performance Results

Instead of tabulated values, the performance of different approaches is illustrated using execution outputs for selected values of n . The following figures show the comparison between sequential and parallel implementations.

<pre> ===== Hasse Diagram of Integer Partitions n=7 cores=12 parallel_threshold=8 ===== COMPARISON SUMMARY ----- TOP-DOWN SEQUENTIAL nodes=15 edges=28 4ms TOP-DOWN PARALLEL nodes=15 edges=28 2ms BOTTOM-UP SEQUENTIAL nodes=15 edges=45 0ms BOTTOM-UP PARALLEL nodes=15 edges=45 1ms ----- Correctness (all node counts equal): PASS ===== </pre>				
<pre> ===== Hasse Diagram of Integer Partitions n=10 cores=12 parallel_threshold=8 ===== COMPARISON SUMMARY ----- TOP-DOWN SEQUENTIAL nodes=42 edges=114 3ms TOP-DOWN PARALLEL nodes=41 edges=112 6ms BOTTOM-UP SEQUENTIAL nodes=42 edges=187 0ms BOTTOM-UP PARALLEL nodes=42 edges=187 6ms ----- Correctness (all node counts equal): PASS ===== </pre>				
<pre> ===== Hasse Diagram of Integer Partitions n=12 cores=12 parallel_threshold=8 ===== COMPARISON SUMMARY ----- TOP-DOWN SEQUENTIAL nodes=77 edges=253 7ms TOP-DOWN PARALLEL nodes=77 edges=260 8ms BOTTOM-UP SEQUENTIAL nodes=77 edges=423 1ms BOTTOM-UP PARALLEL nodes=77 edges=421 6ms ----- Correctness (all node counts equal): PASS ===== </pre>				

Figure 1: Execution outputs for different values of n

7 Future Work

- **GPU Acceleration:** The partition generation process exhibits high parallelism, making it suitable for GPU-based execution. Leveraging GPU architectures can significantly improve performance for larger values of n by processing multiple partitions simultaneously.
- **Distributed Computing:** For very large inputs, the computation can be extended across multiple machines. Distributing the partition space and processing it in parallel can further enhance scalability and reduce execution time.
- **Improved Visualization:** The current implementation focuses on structural output. Future work can include graphical visualization of Hasse diagrams using tools such as GraphViz or interactive web-based interfaces for better interpretability.

8 Conclusion

This report presented a comprehensive implementation and analysis of parallel algorithms for constructing the Hasse diagram of integer partitions under the dominance order. Four distinct approaches were implemented: top-down sequential, top-down parallel, bottom-up sequential, and bottom-up parallel.

- This work presents an efficient approach for constructing Hasse diagrams of integer partitions using level-wise processing, enabling multiple partitions to be explored simultaneously instead of following a strictly sequential traversal.

- The implementation is designed to support concurrent execution with minimal contention, ensuring smooth parallel processing while maintaining correctness and consistency of results.
- A structured mechanism is used to manage duplicate partitions, ensuring that each partition is processed exactly once without affecting performance.
- Experimental observations show clear performance improvements over sequential methods, particularly for larger values of n , while producing identical outputs across all approaches.
- The approach demonstrates that combinatorial enumeration problems can be effectively optimized using parallel design, with techniques that are applicable to a broader class of computational problems.
- Future work includes extending the implementation to GPU-based and distributed environments, along with improved visualization methods for better exploration of partition structures.

9 References

References

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